



TRINITY GRAMMAR SCHOOL
MATHEMATICS DEPARTMENT



YEAR 12 2010 ASSESSMENT TASK 2

MATHEMATICS

(2 UNIT/EXTENSION 1)

Time Allowed – one hour

WEIGHTING 20% towards final result

Outcomes referred to: P2, P4, P5, P6, P7, P8, H1, H2, H4, H5, H6, H7, H8, H9

INSTRUCTIONS:

1. Attempt **ALL** questions.
2. Show all necessary working.
3. **Begin** each question on a **new page**.
4. Each question is of equal value. Mark values are shown besides each part.
5. Non-programmable silent Board of Studies approved calculators are permitted.
6. If requested, additional writing sheets may be obtained from the examinations supervisor upon request.
7. A double sided A4 page of notes is permitted to be referred to throughout this task.

Question 1: Start question on a new page.

(a) Find the primitives of:

(i) 3 **1**

(ii) $x^3 - 3x^2$ **2**

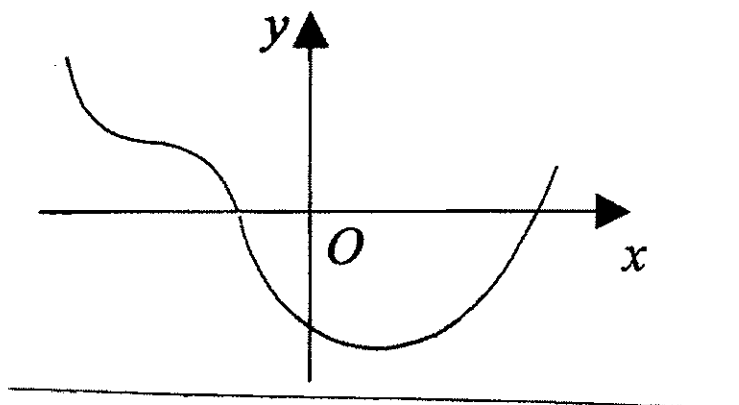
(iii) $(3x + 5)^3$ **2**

(iv) $\frac{x+1}{\sqrt{x}}$ **3**

(b) Evaluate $\int_2^4 (x + \frac{1}{x})^2 dx$ **4**

Question 2: Start question on a new page.

- (a) The graph shows the graph of $y = f(x)$.



- (i) Copy this diagram onto your answer sheet.

- (ii) On the same set of axes, sketch the graph of its derivative, $y = f'(x)$.

3

- (b) Find the equation of the normal to the curve $y = \sqrt{x}$ at the point on the curve where $x = 9$.

4

- (c) For the function $f(x) = x^3 - 3x^2 - 9x + 5$, find the values of x for which:

- (i) the function $f(x)$ is increasing

3

- (ii) the curve $y = f(x)$ is concave up

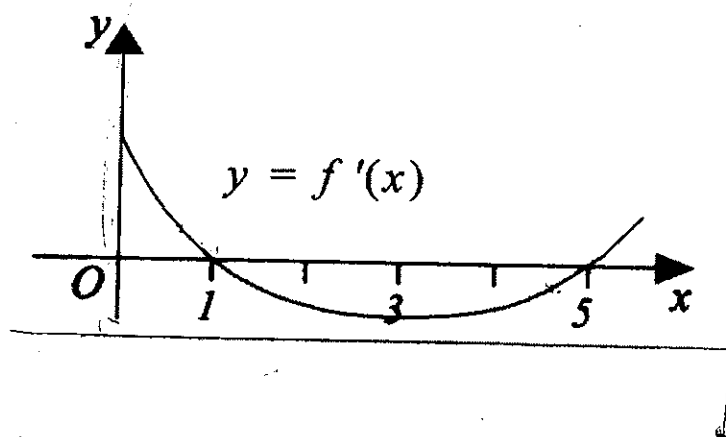
2

Question 3: Start question on a new page.

- (a) The curve $y = 3x^2 + \frac{a}{x^2}$ has a turning point at $x = 2$. Find the constant a .

3

- (b) The diagram shows the graph of a gradient function of the curve $y = f'(x)$.



For what values of x does $f(x)$ have a local maximum? Justify your answer.

2

- (c) The gradient of a curve is given by $\frac{dy}{dx} = 3x^2 - 6x - 9$. The curve passes through the point $(1, -2)$.

- Find the equation of the curve.
- Find the coordinates of the stationary points and determine their nature.
- Find the coordinates of the point of inflexion.
- Sketch the curve in the domain $-2 \leq x \leq 3$ showing its stationary points, point of Inflexion and the y -intercept.

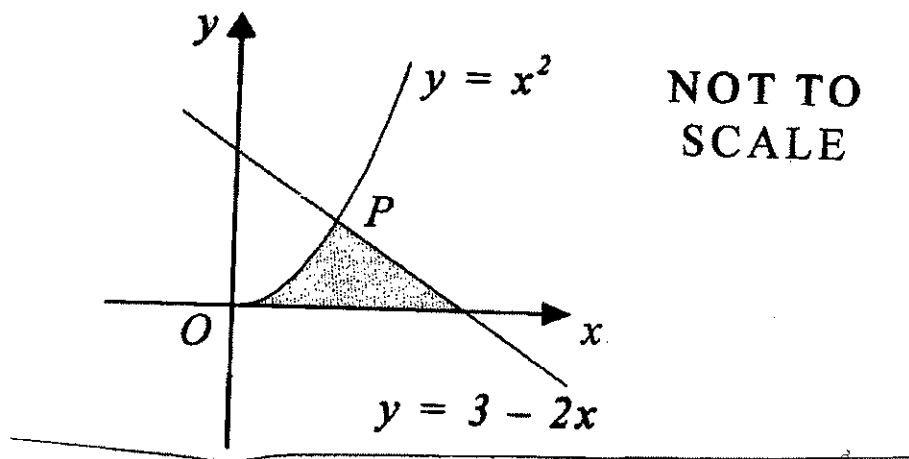
7

Question 4: Start question on a new page.

- (a) (i) Sketch the function $y = x(x^2 - 1)$ on your answer sheet, showing all x -intercepts. 2

- (ii) Find the area bounded by the function $y = x(x^2 - 1)$ and the x -axis. 4

(b)



- (i) Show that the coordinates of the point P are (1, 1). 2
- (ii) Find the volume of the solid formed when the area is rotated about the x -axis. 4

Question 5: Start question on a new page.

- (a) At the point $P(-1, 13)$ on the curve $y = ax^3 + bx^2 + 15x + 26$, there is a point of inflexion.

Find the values of a and b .

4

- (b) A closed water tank in the shape of a right cylinder is to be constructed with a surface area of $54\pi \text{ cm}^2$. The height of the cylinder is $h \text{ cm}$ and the base radius is $r \text{ cm}$.

- (i) Show that the height of the water tank in terms of r is given by $h = \frac{27}{r} - r$.

2

- (ii) Show that the volume V that can be contained in the tank is given by

$$V = 27\pi r - \pi r^3.$$

2

- (iii) Find the radius $r \text{ cm}$ which will give the cylinder its greatest possible volume.

Justify your answer.

4

END OF ASSESSMENT

Solutions.



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Q1 - Wereny
Q2 - Stathalis
Q3 - Chadler
Q4 - Warden
Q5 - Rogerson

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Question 1:

(a) (i) $3x + C$ 1

(ii) $\frac{x^4}{4} - \frac{3x^3}{3} + C$ 2

(iii) $\frac{(3x+5)^4}{3 \times 4} + C$ 2

$= \frac{(3x+5)^4}{12} + C$

(iv) $\int \frac{x}{\sqrt{x}} + \frac{1}{\sqrt{x}} dx$ 1

$= \int x^{\frac{1}{2}} + x^{-\frac{1}{2}} dx$ 1

$= \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} + 2x^{\frac{1}{2}} + C$ 1

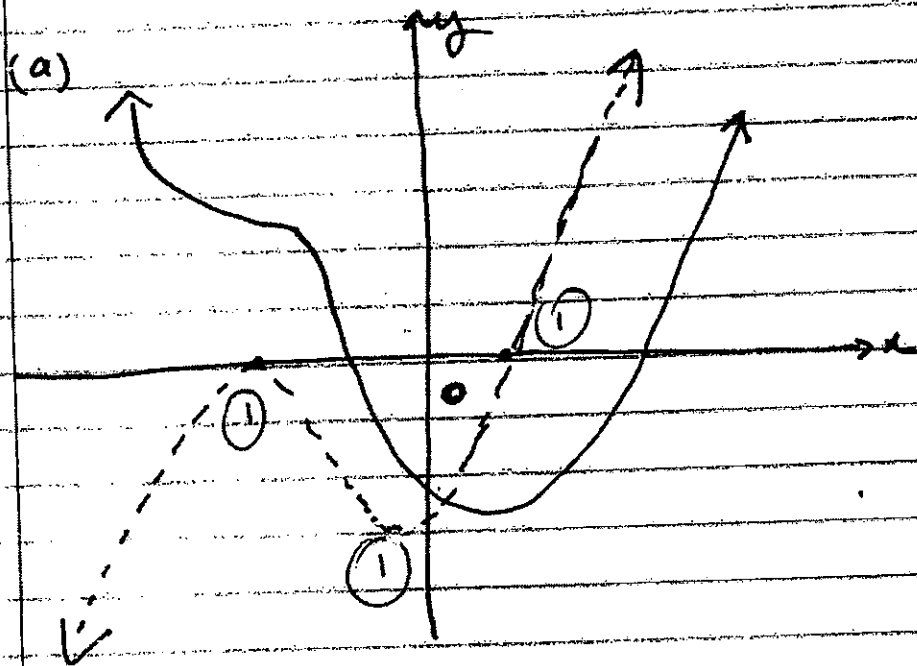
(b) $\int_2^4 \left(x + \frac{1}{x}\right)^2 dx$

$= \int_2^4 (x^2 + 2 + x^{-2}) dx$ 1

$= \left[\frac{x^3}{3} + 2x + \frac{x^{-1}}{-1} \right]_2^4$ 1

$= \left[\frac{x^3}{3} + 2x - \frac{1}{x} \right]_2^4 = \left(\frac{64}{3} + 8 - \frac{1}{4} \right) - \left(\frac{8}{3} + 4 - \frac{1}{2} \right)$
 $= \left(\frac{349}{12} \right) - \left(\frac{37}{6} \right) = \frac{275}{12}$

Question 2:



(b)

$$y = \sqrt{x}$$

$$x=9, y=3$$

$$y = x^{1/2} \quad (1)$$

$$\frac{dy}{dx} = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}$$

$$\therefore m = \frac{1}{6} \quad (1) \quad \therefore m_{\perp} = -6$$

$$y - 3 = -6(x - 9) \quad (1)$$

$$y - 3 = -6x + 54$$

$$y = -6x + 57 \quad (1)$$

$$(c) \quad f(x) = x^3 - 3x^2 - 9x + 5$$

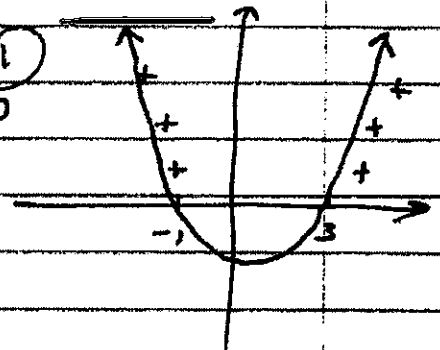
$$(i) \quad f'(x) = 3x^2 - 6x - 9 \quad (1)$$

$$\therefore f'(x) > 0$$

$$3(x^2 - 2x - 3) > 0$$

$$3(x+1)(x-3) > 0 \quad (1)$$

$$\therefore \begin{aligned} x &> 3 \\ x &< -1 \end{aligned} \quad (1)$$



$$(ii) \quad f''(x) = 6x - 6$$

$$\therefore f''(x) > 0 \quad (1)$$

$$6x - 6 > 0$$

$$\begin{aligned} 6x &> 6 \\ x &> 1 \end{aligned} \quad (1)$$

Question 3:

(a) $y' = 6x - 2ax^{-3}$
 $y' = 6x - \frac{2a}{x^3}$ (1)

at $x=2$

$y' = 6(2) - \frac{2a}{8}$ (1)

$0 = 12 - \frac{2a}{8}$

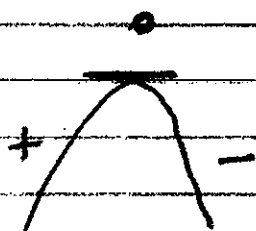
$\frac{2a}{8} = 12$

$2a = 96$

$a = 48$

(1)

(b)



at $x=1$ (1)

$x < 1 \quad \frac{dy}{dx} > 0$

$x > 1 \quad \frac{dy}{dx} < 0$

(1)

$$(c) (i) y = x^3 - 3x^2 - 9x + c$$

$$(1, -2)$$

$$-2 = 1 - 3 - 9 + c$$

$$-2 = -11 + c$$

$$9 = c$$

(1)

$$\therefore y = x^3 - 3x^2 - 9x + 9$$

$$(ii) \quad 0 = 3(x^2 - 2x - 3)$$

$$0 = (x-3)(x+1)$$

$$\textcircled{1} \quad x=3, \quad x=-1$$

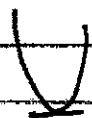
$$y=-18, \quad x=14$$

(1)

$$\frac{dy}{dx^2} = 6x - 6$$

$$\text{At } x=3 \quad \frac{d^2y}{dx^2} = 6(3) - 6$$

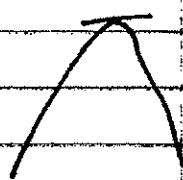
$$= 12$$

 min.

$$\text{At } x=-1 \quad \frac{d^2y}{dx^2} = 6(-1) - 6$$

$$= -6 - 6$$

$$= -12$$

 max.

$$(iii) \quad \frac{d^2y}{dx^2} = 0$$

$$6x - 6 = 0$$

$$6x = 6$$

$$x = 1$$

x	0	1	2	(1)
y''	-6	0	6	

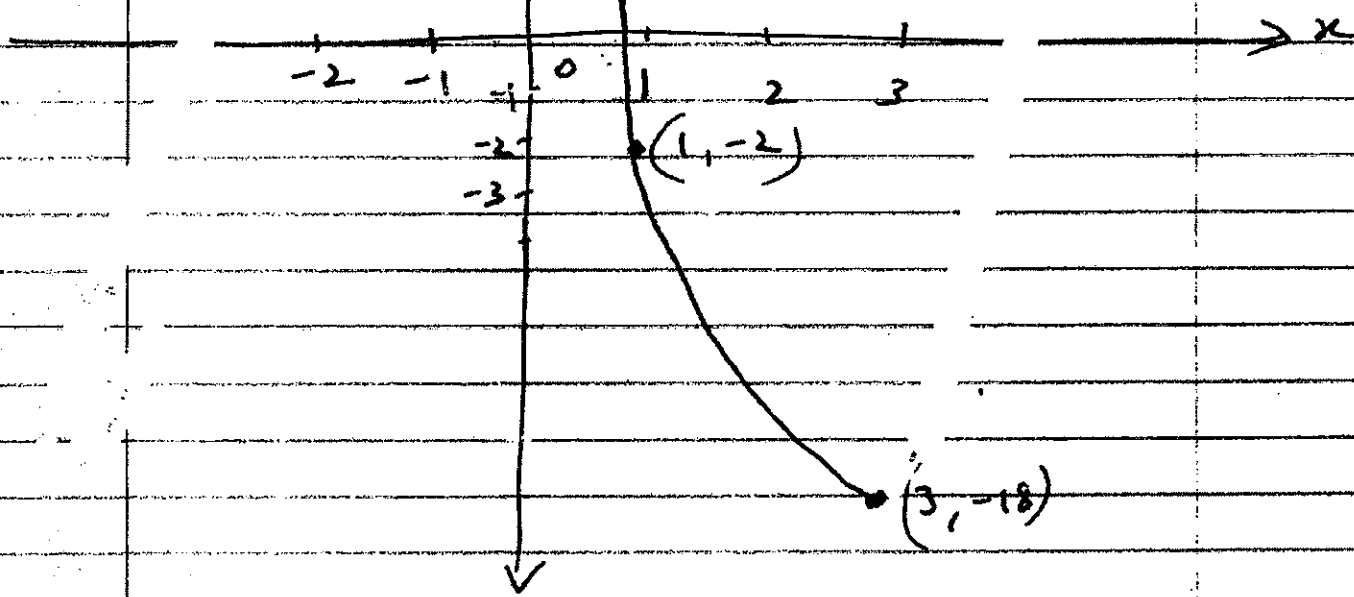
$x = 1, y = -2$ pt of inflexion.

(iv)

$(-1, 14)$
 $(-2, 7)$

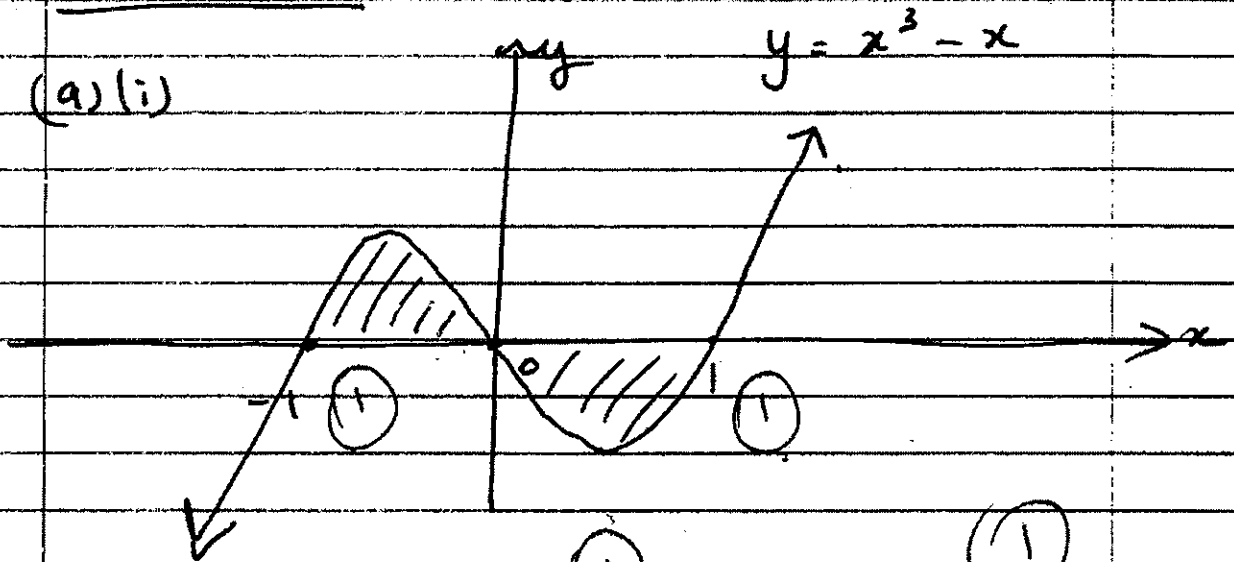
(1)

all points shown.



Question 4:

(a)(i)



(ii) $A = \int_{-1}^0 x^3 - x \, dx + \left| \int_0^1 x^3 - x \, dx \right|$

$$A = \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0 + \left| \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_0^1 \right|$$

$$A = \left[(0) - \left(\frac{1}{4} - \frac{1}{2} \right) \right] + \left| \left(\frac{1}{4} - \frac{1}{2} \right) - (0) \right|$$

$$A = \frac{1}{4} + \frac{1}{4}$$

$$A = \frac{1}{2} \text{ unit}^2$$

$$(b) (i) x^2 = 3 - 2x$$

$$x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0$$

$$\therefore x = -3, x = 1$$

$$\text{if } x = 1, y = 3 - 2(1) = 1$$

$$P(1,1)$$

$$(ii) V = \pi \int_0^1 (x^2)^2 dx + \pi \int_1^{3/2} (3-2x)^2 dx$$

$$V = \pi \int_0^1 x^4 dx + \pi \int_1^{3/2} 9 - 12x + 4x^2 dx$$

$$V = \pi \left[\left(\frac{x^5}{5} \right)_0^1 + \left[9x - 6x^2 + \frac{4x^3}{3} \right]_1^{3/2} \right]$$

$$V = \pi \left[\frac{1}{5} + \left(\left(9 \times \frac{3}{2} \right) - 6 \left(\frac{9}{4} \right) + \frac{4 \left(\frac{27}{8} \right)}{3} - \left(9 - 6 + \frac{4}{3} \right) \right) \right]$$

$$V = \pi \left[\frac{1}{5} + \left[4\frac{1}{2} - \left(4\frac{1}{3} \right) \right] \right]$$

$$V = \pi \left[\frac{1}{5} + \frac{1}{6} \right]$$

$$V = \frac{11\pi}{30} \text{ units}^3$$

Question 5:

(a) $13 = -a + b - 15 + 26$

$$13 = -a + b + 11$$

$$2 = -a + b$$

①

①

$\frac{dy}{dx}$

$$= 3ax^2 + 2bx + 15$$

$\frac{d^2y}{dx^2}$

$$= 6ax + 2b$$

$$0 = -6a + 2b$$

②

①

$$4 = -2a + 2b$$

①'

$$0 = -6a + 2b$$

②

$$\therefore \textcircled{1}' - \textcircled{2}$$

$$4 = 4a$$

$$\boxed{1 = a}$$

①

sub a into

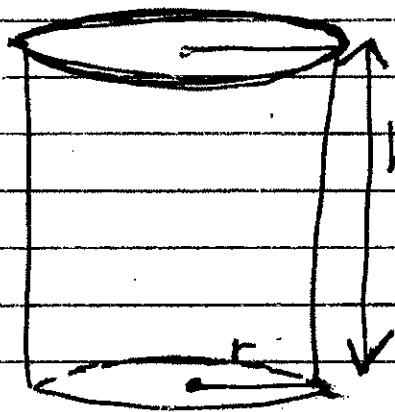
①

$$2 = -1 + b$$

$$\boxed{3 = b}$$

①

(b)



$$SA = 54\pi \text{ cm}^2$$

$$(i) \quad SA = 2\pi r^2 + 2\pi rh$$

(1)

$$54\pi = 2\pi r^2 + 2\pi rh$$

$$\frac{54 - 2r^2}{2r} = h$$

$$\frac{27}{r} - r = h \quad (1)$$

$$(ii) \quad V = \pi r^2 h$$

$$V = \pi \times r^2 \times \left(\frac{27}{r} - r \right)$$

$$V = \frac{27\pi r^2}{r} - \pi r^3$$

(1)

$$V = 27\pi - \pi r^3$$

(1)

(iii)

$$\frac{dV}{dr} = 2\pi - 3\pi r^2 \quad (1)$$

$$0 = 2\pi - 3\pi r^2$$

$$0 = 2 - 3r^2 \quad (1)$$

$$3r^2 = 2$$

$$r^2 = \frac{2}{3}$$

$$r = +\sqrt{\frac{2}{3}}$$

(1)

r	0.5	$\sqrt{\frac{2}{3}}$	1
$\frac{dV}{dr}$	+	0	-

